



Effects of oxygen depletion on soot production, emission and radiative heat transfer in opposed-flow flame spreading over insulated wire in microgravity

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ABSTRACT

The main objective of this article is to investigate experimentally and numerically the effects of reduced-oxygen contents on the soot production and emission from solid fuels in microgravity, which constitute an important issue in terms of fire safety for manned space missions. Due to its convenience for the implementation of soot-related optical diagnostics, the configuration of a flame spreading in an opposed flow over thin nickel chromium (NiCr) wires coated by low density PolyEthylene (LDPE) is considered. Experiments are conducted at a pressure of 101.3 kPa and an oxidizer velocity of 150 mm/s. The oxygen mole fraction in the oxidizer, X_{O_2} , is varied from 18% to 21% by nitrogen dilution of air. The modeling strategy lies on a surrogate fuel to mimic the combustion of LDPE by preserving its stoichiometry and the laminar smoke-point (LSP) flame height. The numerical model considers a detailed chemistry, a two-equation soot production model involving laminar smoke point (LSP)-based soot formation rates and oxidation by OH and O_2 , a radiation model coupling the Full-Spectrum correlated-k method with the finite volume method, and a simple degradation model for LDPE. Based on experimental evidence, the soot formation rate is scaled by the adiabatic flame temperature to account for thermal effects due to variation in X_{O_2} . The model reproduces quantitatively the increase in flame size, residence time, and soot volume fraction observed experimentally as X_{O_2} is enhanced as well as the transition from a non-smoking to a smoking flame, which occurs for X_{O_2} between 19% and 20%. The increase in soot volume fraction results of a combined enhancement in both residence time, owing to an increase in the fuel mass flow rate, and soot formation rate due to higher temperature in the soot formation region. The radiant fraction increases significantly with X_{O_2} from about 17% for $X_{O_2}=18\%$ to about 36% for $X_{O_2}=21\%$. This increase in radiative losses is accompanied by a reduction of the temperature in the soot oxidation region. Therefore, for increasing X_{O_2} , the soot oxidation process is governed by a competition between oxygen-enhanced conditions that promote the formation of soot oxidizing species and the increase in radiative losses that dampens their formation. For the present flames, the first mechanism prevails as X_{O_2} increases from 18% to 19% whereas the second dominates as X_{O_2} is further increased, leading to smoking flames as actually observed for $X_{O_2}=20$ and 21%. The radiant fraction at the smoke-point transition and the soot oxidation freezing temperature are in line with those reported at normal gravity. Finally, model results show that, whatever X_{O_2} , the contribution of radiation to the heating process is negligible ahead of the pyrolysis front and is largely overcome by surface radiative losses along the pyrolysis region.

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1. Introduction

Ignition and flame spread over idealized thin electrical wires, composed by a metal core coated with a polymer, has received a considerable attention over the last twenty years due to the high interest of this configuration for fire safety [1]. In particular, elec-

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Nomenclature

A_S	soot surface area [m^{-1}]
E_a	activation energy [kJ/mol]
f_S	soot volume fraction [-]
L_{fl}	flame length [m]
L_S	distance from the molten droplet leading edge to the integrated SVF peak [m]
l_{sp}	laminar smoke point height [m]
$\dot{m}_p$	pyrolysis mass flow rate [$\text{kg}\cdot\text{s}^{-1}$]
$\dot{m}_p''$	pyrolysis mass flow rate per unit area [$\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$]
N_A	Avogadro number [$\text{part}\cdot\text{mol}^{-1}$]
NC_{min}	number of carbon atoms in the incipient soot particle [-]
N_S	soot number density per unit mass of mixture [$\text{part}\cdot\text{kg}^{-1}$]
P	pressure [Pa]
P_{OH}	OH partial pressure [Pa]
P_{O_2}	O_2 partial pressure [Pa]
$\dot{q}_{R,inc}''$	incident radiative flux [$\text{W}\cdot\text{m}^{-2}$]
$\dot{q}_{R,w}''$	surface re-radiation [$\text{W}\cdot\text{m}^{-2}$]
r	radial coordinate [m]
r_{fl}	flame radius [m]
R_{wire}	wire radius [m]
R_u	universal gas constant ($\text{kJ}/(\text{mol K})$)
S	Stoichiometric oxidizer/fuel volumetric ratio [-]
s	Stoichiometric oxidizer/fuel mass ratio [-]
S_b	surface of the molten ball [m^2]
T	temperature [K]
T_{ad}	adiabatic flame temperature [K]
T_{ad}^0	adiabatic flame temperature for $X_{O_2} = 20\%$ [K]
u_F	Fuel injection velocity [$\text{m}\cdot\text{s}^{-1}$]
u_p	spread rate [$\text{m}\cdot\text{s}^{-1}$]
u_{ox}	oxidizer flow velocity [$\text{m}\cdot\text{s}^{-1}$]
W_i	molecular weight of the i^{th} species [$\text{kg}\cdot\text{mol}^{-1}$]
x_i	mole fraction of the i^{th} species [-]
X_{O_2}	mole fraction of oxygen in the oxidizer [-]
z	axial coordinate [m]
α_g	gas thermal diffusivity [$\text{m}^2\cdot\text{s}^{-1}$]
δ	stand-off distance [m]
η_{OH}	collision efficiency of OH [-]
ξ	mixture fraction [-]
ρ_g	gas density [$\text{kg}\cdot\text{m}^{-3}$]
τ	residence time [s]
χ_R	radiant fraction [-]
$\dot{\omega}_{coag}$	reaction rate for soot coagulation [$\text{part}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_n$	reaction rate for soot nucleation [$\text{mol}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{N_S}$	reaction rate for soot number density [$\text{part}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{O_2}''$	reaction rate for soot oxidation by O_2 per unit soot surface area [$\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$]
$\dot{\omega}_{OH}''$	reaction rate for soot oxidation by OH per unit soot surface area [$\text{kg}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$]
$\dot{\omega}_{OH}$	reaction rate for soot oxidation by OH [$\text{kg}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{SF}$	reaction rate for soot formation rate [$\text{kg}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{SF,P}^{X_{O_2}^0}$	peak of soot formation rate for $X_{O_2}^0=20\%$ [$\text{kg}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{SF,P}^{X_{O_2}}$	peak of soot formation rate for X_{O_2} [$\text{kg}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]
$\dot{\omega}_{Y_S}$	source term for soot mass fraction [$\text{kg}\cdot\text{m}^{-3}\cdot\text{s}^{-1}$]

Subscript

ad adiabatic

b	molten ball
fl	flame
g	gas
inc	incident
max	maximum
p	pyrolysis
PE	polyethylene
R	radiation or radiative
sp	smoke point
S	soot
∞	ambient

Superscript

0 reference

Acronyms

BMAE	Broadband Modulated Absorption Emission
FSCK	Full-Spectrum correlated-k
LDPE	low density polyethylene
LSP	laminar smoke point
PE	polyethylene
SF	soot formation
SP	smoke point
SVF	soot volume fraction
SO	soot oxidation

trical wires were identified as the primary cause of fire initiation and growth in a space vehicle. This has resulted in specific experimental, theoretical and numerical studies in microgravity environments to get insights into flame spread characteristics and flammability properties [2–12].

The release of smoke is a critical aspect of fire safety, as evidenced by the various fire incidents reported in the context of manned space missions [13–18]. In addition, smoke and flame radiation constitute signatures of fire appearance to develop fire detection strategies in microgravity conditions [2]. Although most of these incidents, at the noticeable exception of that aboard Mir in 1997, involved smoke released in the absence of flame, and the early detection of fires from smoke generated by oxidative pyrolysis (such as smoldering) is desirable to allow the crew to respond before a larger flaming situation develops [19], the soot released from flaming solids, including polymers, is also of fundamental and applicative interest [20,21]. Beyond the study of material flammability, the opposed flow spreading of laminar diffusion flames over idealized cylindrical electrical wires in microgravity offers a unique configuration to investigate soot production, smoke release and, corresponding radiative heat transfer processes from flaming solid fuels [21–26]. The 2D axisymmetric geometry enables the implementation of advanced optical diagnostics based on extinction techniques, such as the BMAE (Broadband Modulated Absorption Emission) technique [27], to measure local soot volume fraction (SVF) and temperature. A proper selection of the metallic core enables the formation of a stable laminar boundary layer diffusion flame spreading at a steady rate over the 22 s of microgravity in parabolic flights [22–26]. In addition, the presence of the metallic core, on the one hand, ensures the rigidity of the configuration maintaining an axisymmetric boundary layer within the flame spread and, on the other hand, avoids the occurrence of a dripping process. Finally, the microgravity environment promotes longer residence times than those encountered in buoyant flames. The effects of ambient conditions, namely pressure, P , oxygen content in the oxidizer, X_{O_2} , and oxidizer flow velocity, u_{ox} , on soot production and emission were experimentally investigated using the BMAE in parabolic flights for flames propagating in opposed flow over thin electrical wires. The studied samples were

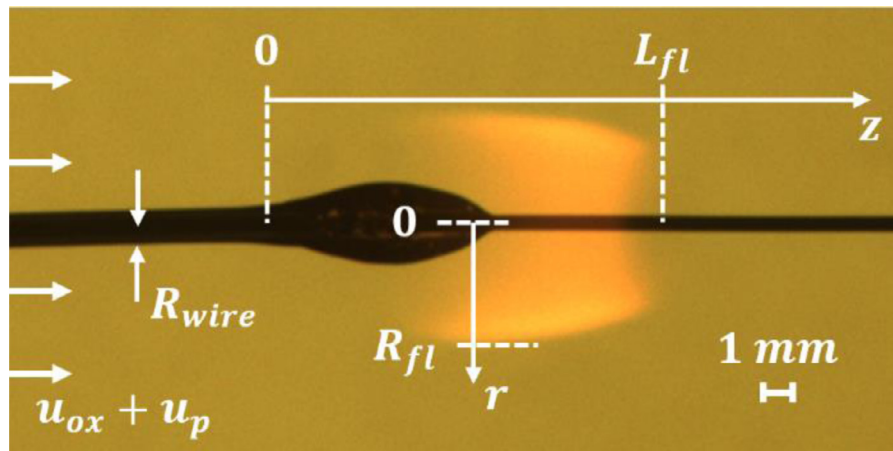


Fig. 1. Backlighting frame imaging a flame spreading over a wire in opposed flow conditions in microgravity. Main features of the flame geometry, subsequent notations adopted, and a 1 mm scale are overlaid.

made of a 0.5 mm in diameter NiCr metallic core coated by a 0.3 mm thick layer of LDPE [23,24]. P , X_{O_2} , and u_{ox} were varied between 51 kPa and 142 kPa, 18% and 21%, 100 mm/s and 200 mm/s, respectively. LDPE was selected as solid fuel for three main reasons. First, it corresponds to an international target configuration to investigate fundamentally the flammability properties of electrical wires [2,4,8–12,22–26]. Second, it produces SVF levels above the detection threshold of the BMAE over the ranges of pressure, oxygen content, and flow velocity considered. Finally, the corresponding flames undergo a transition from a non-smoking flame to a smoking one over these ranges.

These experimental investigations were accompanied by the development of a numerical model, involving the coupling of a detailed chemistry and a state-of-the-art radiation model, to help in the interpretation of data [25,26]. The main challenge in the modeling of the gas-phase processes resulting from the combustion of solid fuels lies in the fact that the gaseous pyrolysis products are generally not well identified and some simplifications, such as a global one-step chemical reaction, have to be introduced due to this lack of information. In order to circumvent this limitation and to be able to consider a detailed chemistry with access to soot precursors and soot oxidizing species such OH radicals, a numerical fuel surrogate, inspired from the works of Refs. [28,29], was designed to mimic the gas-phase combustion of PE in terms of flame structure and fuel sooting propensity. In our approach the fuel sooting propensity is related to the laminar smoke-point (LSP) height and the LSP strategy, initiated in Refs. [30,31], was adopted to determine the soot production rates of PE. This strategy is based on the analysis developed by De Ris et al. [32] from experimental evidence that the soot production rates of a given fuel are inversely proportional to the LSP height. In practice, the soot production rates are established for a reference fuel, typically ethylene, and the model is extended to other fuels by scaling the aforementioned rates by the inverse of the corresponding LSP flame height [30,31]. In our approach, a two-equation LSP-based soot production model was specifically developed, extending the one-equation formulation proposed in Refs [30,31]. This extended approach is expected to improve the soot oxidation description by considering explicitly the surface area dependence of the soot oxidation rates by O_2 and OH [33]. The combination of the experimental results and simulations provided a detailed characterization of the effects of pressure on flame structure, soot production and soot emission [23,26]. More specifically, it was found that increasing the pressure does not affect the residence time in the present configuration and that, over the range of pressures investigated, the soot formation rate of LDPE is third order in pressure. In addition, con-

sistently with data obtained at normal gravity [34–37], the smoke-point transition was found to occur for a radiant fraction of about 0.3 and the soot oxidation freezing temperature was estimated in the range 1350–1450 K.

The present article focuses on the effects of X_{O_2} on flame structure, soot production, and soot emission in the opposed flow flame spreading over NiCr/LDPE wires for X_{O_2} lower than 21%. The main objectives are: i) to extend the previously-developed two-equation LSP-based soot production model to account for X_{O_2} effects, ii) to further validate the surrogate fuel modeling strategy by comparison with the available experimental data reported in Refs. [23,24], and iii) to provide insights into the ways X_{O_2} affects both soot formation (SF) and soot oxidation (SO) processes as well as the resulting smoke emission and radiative heat transfer. The article is organized as follows. Sections 2 and 3 present the experimental set-up and the main features of the numerical model, respectively. The two-equation LSP model and the developments performed to account for X_{O_2} effects are described in Section 4. The results are discussed in Section 5. Finally, Section 6 summarizes the main conclusions of the paper.

2. Experimental set-up

Experiments were conducted in parabolic flights. The experimental setup and the optical diagnostics are detailed in Refs. [22–24] and are only briefly described here. Cylindrical 150 mm long wires, consisting of a 0.5 mm diameter NiCr core coated with a 0.3 mm thick LDPE insulation, are placed along the central axis of a cylindrical combustion chamber with an inner diameter of 190 mm. For all the experiments considered in the present study, the pressure and flow velocity were set equal to 101.3 kPa and 150 mm.s⁻¹, respectively. X_{O_2} was varied between 18% and 21% by nitrogen dilution of air. The samples were ignited using a hot Kanthal wire at the beginning of the parabolic sequence, resulting in an opposed-flow flame spread. As discussed in the introduction, this choice of NiCr and dimensions enable the formation of a stable steady-spreading axisymmetric laminar boundary layer diffusion flame over the 22 s of microgravity in parabolic flights for the range of ambient conditions studied [22–26].

Flame spread rate, pyrolysis mass flow rate, and SVF were determined during the steady flame spread using the optical diagnostics described in details in Ref. [23]. In particular, SVF fields were obtained using the B-MAE technique as extensively outlined in Ref. [27]. This technique enables the recovery of the SVF and temperature with an uncertainty lower than 1 ppm and 100 K, respectively. In agreement with observations reported in previous

Table 1

Main characteristics of the flames investigated in the present study. The pressure and the oxidizer velocity are 101.3 kPa and 150 mm.s⁻¹, respectively.

(%)	18	19	20	21
u_p (mm/s)	1.11 ± 0.08	1.27 ± 0.07	1.33 ± 0.01	1.46 ± 0.07
$\dot{m}_p$ (mg/s)	0.77 ± 0.058	0.879 ± 0.048	0.925 ± 0.01	1.01 ± 0.05
L_f (mm)	12.41 ± 0.42	14.40 ± 0.77	15.50 ± 0.36	16.19 ± 0.32
r_f (mm)	3.26 ± 0.06	3.47 ± 0.19	3.62 ± 0.28	3.78 ± 0.08
$f_{S,max}$ (ppm)	10.20 ± 0.82	13.85 ± 1.12	16.73 ± 0.96	22.70 ± 1.36
L_S (mm)	10.01 ± 0.21	11.40 ± 0.31	12.40 ± 0.24	13.50 ± 0.12
τ_{SF} (ms)	66.60	76.00	82.66	90.00
S (-)	16.67	15.78	15.00	14.28
Smoking/non-smoking	Non-smoking	Non-smoking	Smoking	Smoking

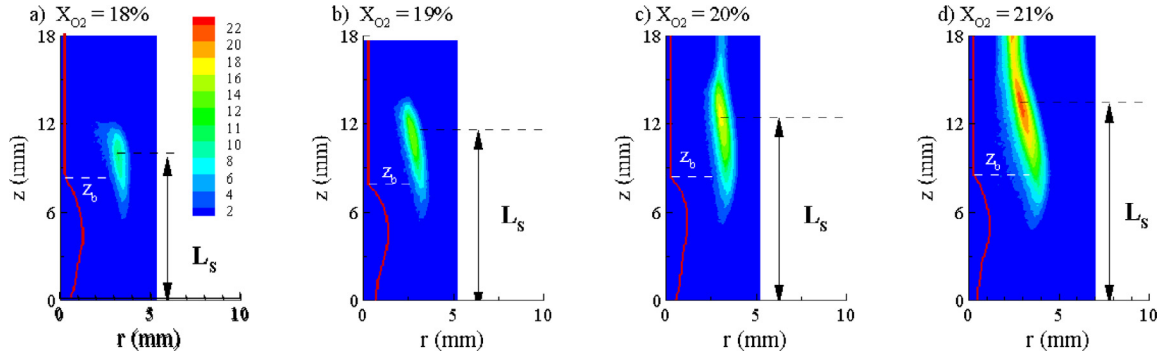


Fig. 2. Fields of SVF in ppm for a) $X_{O_2}=18\%$, b) $X_{O_2}=19\%$, c) $X_{O_2}=20\%$ and d) $X_{O_2}=21\%$. The red solid line represents the wire surface. The origin of the z -coordinate is located at the molten ball leading edge that is assumed to correspond to the pyrolysis front. z_b represents the trailing edge of the pyrolyzing region. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

works [8,21,26,38,39], LDPE was characterized by the formation of gaseous bubbles in the molten phase as it degrades, which then burst and eject fragments of molten PE into the gas phase after they reached the fuel surface. These events were mainly observed at higher oxygen content ($X_{O_2}=0.21$) and were found to be sufficiently sparse to affect significantly neither the mean soot production level nor the smoke point transition at the flame trailing edge. Nevertheless, although the resulting perturbations do not affect the flame downstream, the local signal deformation in flame emission means that the error on local flame temperature increases drastically [26]. As a result, only SVF experimental fields are reported here.

The flame was assumed to spread at a steady rate once the dimensions of the characteristic molten insulation droplet (see Fig. 1), the visible flame length, and the rate of the flame front displacement reach a steady state value [22]. The flame spread rate, u_p , is evaluated by tracking the time evolution of the luminous flame front whereas the pyrolysis mass flow rate, $\dot{m}_p$, is then evaluated from u_p assuming $\rho_{PE} = 920 \text{ kg/m}^3$.

Table 1 presents the main characteristics of the four flames investigated in the present study. As illustrated on Fig. 1, the luminous flame radius, r_f , and flame length, L_f , are defined as the distances between the coated wire and the maximum radial location of the luminous flame and between the pyrolysis front, identified by molten droplet leading edge, and the maximum axial location of the luminous flame, respectively. Similarly, the SF characteristic length scale, L_S , is defined as the axial distance between the molten droplet leading edge and the peak of SVF (see Fig. 2). Table 1 shows that u_p , $\dot{m}_p$, L_f , r_f , L_S , and the peak of SVF, $f_{S,max}$, increase with X_{O_2} . $f_{S,max}$ increases from about 10.20 ppm for $X_{O_2}=18\%$ to about 22.70 ppm for $X_{O_2}=21\%$. The residence time for the SF process, that can be estimated as $\tau_{SF} = L_S/u_{ox}$, increases also with X_{O_2} . The flames for $X_{O_2}=18\%$ and 19% are non-smoking whereas the flames for $X_{O_2}=20\%$ and 21% are smoking, showing that the smoke point transition occurs somewhere between $X_{O_2}=19\%$ and 20% (see Fig. 2).

3. Numerical model

A detailed description of the numerical model can be found in Refs. [25,26] and only a summary is given here. The model solves the governing equations of both gaseous and solid phases in a flame-fixed axisymmetric coordinate system.

For the gas phase, the Navier-Stokes equations and transport equations for gas-phase species mass fractions and energy are solved. The numerical methods as well as the strategy for handling the stiffness of the equation system can be found in Refs. [25,26]. As discussed in the introduction, a surrogate fuel is designed to mimic the gas-phase combustion of PE in terms of flame structure and fuel sooting propensity. This strategy, widely inspired from that used to develop the Burning Rate Emulator (BRE) to simulate the burning of condensed fuels [28,29], consists in preserving the heat of combustion (or similarly the oxygen-fuel mass ratio) to reproduce the flame structure, and the laminar smoke point (LSP) height to reproduce the fuel sooting propensity. Pure ethylene is adopted to represent the pyrolysis products of LDPE since both species exhibit very similar heat of combustion with a discrepancy lower than 10% and the same oxygen-fuel mass ratio [40]. The LSP height of LDPE will be preserved in the soot production model in order to mimic its sooting propensity (see Section 4). This surrogate fuel modeling strategy was found to be able to reproduce accurately the experimental flame structure, soot production and soot emission of flames spreading over LDPE wires under different pressure levels [26]. The oxidation of ethylene is modelled using the full chemical kinetic scheme developed by Qin et al. [41].

The present study focuses on gas-phase processes. Therefore, the conjugated heat transfer problem at the gas/solid interface is simplified since its solution is only required to provide proper boundary conditions for the gas-phase modeling. The spread rates, the pyrolysis mass flow rate, the structure of the burning wire and the shapes of the molten LDPE droplet are specified as inputs from the experimental steady spreading data. The steady structure of the burning wire consists of three regions as shown in

Figs. 1 and 2:i) the insulated wire located ahead of the pyrolysis front for $z \leq 0$, ii) the molten ball corresponding to the pyrolysing region for $0 \leq z \leq z_b$, and iii) the bare NiCr core for $z > z_b$. The experimental data in Fig. 2 show that the shape of the molten ball is hardly affected by the different X_{O_2} and, for the sake of simplicity, the molten ball obtained for $X_{O_2} = 19\%$ will be adopted for all the simulations. In the pyrolysing region, the fuel injection velocity profile, $u_F(z)$, is determined from the measured fuel mass flow rate, $\dot{m}_p$, the ball surface, S_b , and the stand-off distance, $\delta(z)$ by assuming that $\dot{m}_p''$ is inversely proportional to $\delta(z)$ [25,26].

The axisymmetric heat transfer equation in the condensed phases is solved for both the metallic core and LDPE by assuming that both LDPE melting and pyrolysis processes behave as phase transitions. Based on these assumptions, the PE located ahead of the pyrolysis front can be in: i) a virgin state if $T_{PE} < T_{melt}$, with $T_{melt} = 403$ K, ii) at the transition if $T_{PE} = T_{melt}$ and, iii) in the molten phase if $T_{PE} > T_{melt}$. The pyrolysis process is assumed to occur at the molten ball surface at a fixed temperature, $T_p = 760$ K. All the thermal constants for the LDPE, including density, conductivity and heat capacity of both virgin and molten LDPE, temperature and heat of melting and pyrolysis temperature were taken from thermal analysis [9,25,42]. During the melting stage, the thermal properties are computed using a mass-weighted average between the properties of solid and those of the molten LDPE [25,26].

A detailed description of the radiation model can be found in Refs. [25,26]. CO_2 , H_2O and soot are considered as the only radiating species since it was found that the contribution of CO is negligible. The radiation model combines the Full-Spectrum Correlated-K (FSCK) [43] as a gas-soot radiative property model and the Finite Volume Method [44] as radiative transfer equation solver. The angular mesh consists in the present simulations of 12×16 control angles. The emissivities of the LDPE, whether in its virgin or molten states, and of the NiCr along the bare wire were set equal to 1.

4. Soot production model

4.1. Formulation

Smoke production is assumed to be only a gas-phase process since, as discussed in Section 2, the generation of solid particles from the bursting of LDPE bubbles were observed mainly for $X_{O_2} = 0.21$ and were found to be sparse enough to have no significant effects on soot production and smoke point transition.

The two-equation LSP soot production model, proposed in Ref. [26] and based on transport equations for soot number density per unit mass of mixture (N_s) and soot mass fraction (Y_s), is adopted and further developed to account for the influence of oxygen depletion on soot formation. Thermophoretic velocities are included in N_s and Y_s conservation equations.

The source term, $\dot{\omega}_s = \dot{\omega}_{SF} - (\dot{\omega}_{O_2}'' + \dot{\omega}_{OH}'')A_s$, accounts for SF and SO by O_2 and OH. The SO rates by OH and O_2 , optimized by Guo et al. [33], were considered:

$$\dot{\omega}_{OH}'' = 1.27 \times 10^{-2} \frac{\eta_{OH} P_{OH}}{\sqrt{T}} \quad (1)$$

$$\dot{\omega}_{O_2}'' = 15.8 \frac{P_{O_2}}{\sqrt{T}} \exp(-E_{A,O_2}/R_u T) \quad (2)$$

with $\eta_{OH} = 0.1$ and $E_{A,O_2} = 195$ kJ/mol [33]. The source term, $\dot{\omega}_{N_s}$, for the soot number density is given by $\dot{\omega}_{N_s} = N_A/NC_{min}\dot{\omega}_n - \dot{\omega}_{coag}$ where N_A and $NC_{min} = 60$ are the Avogadro number and the number of carbon atoms in the incipient soot particle [45], respectively. The coagulation rate, $\dot{\omega}_{coag}$, is computed as proposed by Lindstedt [45] with a Van der Waals enhancement factor of 9.

4.2. Soot formation and nucleation rates

4.2.1. Background

The modeling of the SF rate, $\dot{\omega}_{SF}$, in the case of the combustion of solid fuels is challenging since, as discussed in the introduction, the gaseous fuel released by the material pyrolysis is generally not well identified. In order to circumvent this issue, LSP-based soot production models have emerged over the last decade [30]. The main idea of these models is to map $\dot{\omega}_{SF}$ as a function of mixture fraction, ξ , and temperature, T , for a reference gaseous fuel, typically ethylene, burning in air under atmospheric pressure. The mapping proposed by Lautenberger et al. [30] for ethylene is adopted, leading to $\dot{\omega}_{SF} = f_{SF}(\xi)g_{SF}(T)$. f_{SF} and g_{SF} are cubic functions whose coefficients are determined by specifying the value and slope of the polynomials at the peak and at the lower and higher limits of the SF process.

Based on experimental evidence, the model is extended to other fuels by scaling the peak of SF rate obtained for the reference fuel by the inverse of the corresponding LSP flame height. The LSP flame height quantifies the propensity of a given fuel to release soot [32]. It represents the height of a well-ventilated axisymmetric laminar diffusion flame burning in air under atmospheric pressure at which soot starts to be released from the flame tip [32]. Also based on experimental evidence, LSP-based soot production models can be extended to sub- and super-atmospheric pressures by scaling the peak of SF rate by $(P/P_0)^n$, where P_0 is the atmospheric pressure [32].

Based on these considerations and on the initial formulation of Lautenberger et al. [30], the expression of the peak of soot formation rates can be estimated as:

$$\dot{\omega}_{SF,P} = 1.1 Y_{FT} \frac{l_{sp,C_2H_4}}{l_{sp,PE}} \left(\frac{W_{C_2H_4}}{W_{PE}} \right) \left(\frac{P}{P_0} \right)^n \quad (3)$$

where l_{sp,C_2H_4} and $l_{sp,PE}$ are the SP flame heights of ethylene and PE that have been estimated to be equal to 0.106 m and 0.045 m, respectively [40]. Y_{FT} is the mass fraction of fuel in the fuel supply stream. The constant 1.1 was calibrated to match SF in axisymmetric laminar ethylene diffusion flames burning in air at atmospheric pressure [30].

Eq. (3) was updated to model SF of PE in our previous study [26] for two reasons: i) the composition of the pyrolysis products and, in turn, W_{PE} are not known, and ii) the modeling has considered LDPE samples burning over a range of pressure in under-oxygenated conditions with $X_{O_2} = 20\%$. This particular value of X_{O_2} was the most suitable to investigate pressure effects over soot production since SVF was found to be above the detection threshold of the BMAE over a wide range of pressures while avoiding undesirable instabilities related to the formation and burst of bubbles in the LDPE as mainly observed for $X_{O_2} = 0.21$. As a consequence, the term $(\frac{W_{C_2H_4}}{W_{PE}})$ was ignored in our formulation and the constant 1.1 was recalibrated to match the experimental SVF peak in a flame spreading over a wire coated by LDPE at a nominal pressure $P = 101.3$ kPa, and $X_{O_2} = 20\%$. The calibrated constant was found equal to 0.5, leading to:

$$\dot{\omega}_{SF,P}^{X_{O_2}} = 0.5 Y_{FT} \frac{l_{sp,C_2H_4}}{l_{sp,PE}} \left(\frac{P}{P_0} \right)^n \quad (4)$$

where $X_{O_2} = 20\%$. In addition, an analysis of our experimental data showed that n is equal to 3, approximately, for pressure ranging from 50 kPa to 140 kPa [23,26].

The nucleation rate, $\dot{\omega}_n$, was computed by revisiting the acetylene-based inception model developed by Lindstedt [45]. The initial formulation was multiplied by $l_{sp,C_2H_4}/l_{sp,PE}$ to account for the difference in sooting propensity between ethylene and PE:

$$\dot{\omega}_n = 2k_1(T)l_{sp,C_2H_4}/l_{sp,PE}[C_2H_2] \quad (5)$$

Table 2

Experimental flow conditions for the axisymmetric laminar coflow ethylene diffusion flames at normal gravity and atmospheric pressure.

Ref.	Characteristic	Fuel Stream		Oxidizer Stream	S (-)	T_{ad} (K)
Sun et al. [46]	$d_F=10.5$ mm $\dot{Q}_F=3.85$ cm ³ /s	X_F	X_{N_2}	X_{O_2} (%)		
		1	0	21–36.9	14.28–8.13	2368–2773
		1	0	16.8–0.21	17.85–14.28	2145–2368
		0.8–0.95	0.05–0.20	16.8–0.20	14.28	2130–2317
Zeng et al. [49]	$d_F=16.4$ mm $\dot{Q}_F=2.80$ cm ³ /s	1	0	15.2–20.9	19.73–14.35	2026–2361
Fuentes et al. [47]	$d_F=10.0$ mm $\dot{Q}_F=2.35$ cm ³ /s	1	0	17–35	17.65–8.57	2152–2744

where $[C_2H_2]$ is the acetylene concentration. The kinetic parameter, $k_1(T)$, is provided by Lindstedt [45].

The semi-empirical soot production model described above was found to capture quantitatively the pressure effects on both SF and SO processes for flames spreading over LDPE wires for $X_{O_2}^0=20\%$ [26].

4.2.2. Effects of oxygen depletion

The objective of this section is to extend the present LSP model to account for variations of X_{O_2} with a special emphasis for oxygen-reduced oxidizers. Varying X_{O_2} modifies two fundamental properties of diffusion flames, namely, the volumetric stoichiometric oxidizer to fuel ratio, S , and the adiabatic flame temperature, T_{ad} [46]. Experiments and detailed numerical simulations of axisymmetric laminar coflow diffusion flames at normal gravity suggest that S controls mainly the flame geometry and, in turn, the residence times whereas the effects on SF and SO rates are rather related to thermal effects that affect the formation of soot precursors and soot oxidizing species [31,46–48]. These effects are directly included in soot nucleation (see Eq. (5)) and oxidation rates (see Eqs. (1) and (2)) that depend explicitly on temperature and concentrations of the corresponding species. In addition, the present model assumes that the effects of X_{O_2} on the peak of SF rate are only related to T_{ad} :

$$\dot{\omega}_{SF,P}^{X_{O_2}} = \dot{\omega}_{SF,P}^{X_{O_2}^0} \times \left(\frac{T_{ad}}{T_{ad}^0} \right)^m \quad (6)$$

where T_{ad}^0 is the adiabatic flame temperature for $X_{O_2}^0=20\%$.

The exponent m is determined through the analysis of available experimental data. The selected set of experimental data includes axisymmetric laminar ethylene diffusion flames at normal gravity as well as spreading flames from the present study. The flow conditions relative to the experiments at micro and normal gravity are summarized in Tables 1 and 2, respectively. Table 2 shows that Sun et al. [46] considered a Santoro-type burner with an inner diameter of 10.5 mm for the central fuel tube and an inner diameter of 97.7 mm for the outer tube. For all the experiments, the fuel and oxidizer injection velocities were maintained unchanged and equal to 3.98 cm/s and 24.0 cm/s, respectively, leading to a volumetric fuel flow rate of 3.85 cm³/s. Their comprehensive set of flow conditions included ethylene flames burning in both oxygen-enriched and oxygen-reduced oxidizers. For the oxygen-reduced oxidizers, two sub-sets of experiments were considered. In the first, pure ethylene injected through the central tube while the oxidizer was diluted by nitrogen, leading to variations in S . In the second, both fuel and oxidizer were diluted by nitrogen to maintain S constant. Therefore, for a given X_{O_2} , these two sub-sets of data are characterized by different values of S but very similar values of T_{ad} . This enables separate investigation of the effects of S and T_{ad} on the SF rates. In the experiments of Sun et al. [46], SVF were measured with planar Laser Induced Incandescence (LII). Zeng et al. [49] also used LII technique to measure SVF in laminar axisymmetric ethylene-air/N₂ diffusion flames generated from a Santoro type burner with an inner diameter for the central fuel tube of 16.4 mm. The fuel and oxidizer flow rates were maintained at 2.80

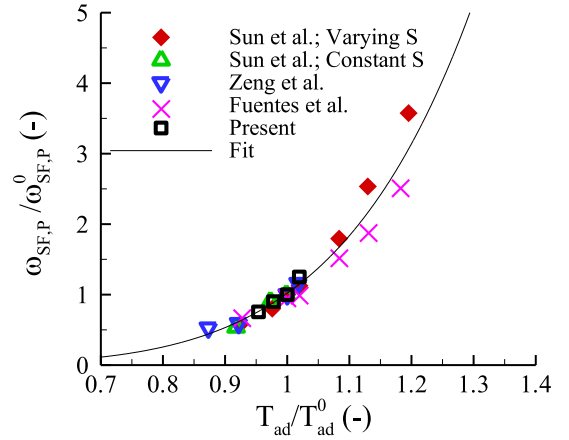


Fig. 3. $\dot{\omega}_{SF,P} / \dot{\omega}_{SF,P}^0$ as a function of T_{ad} / T_{ad}^0 . This figure includes the experimental data obtained from Sun et al. [46], Zeng et al. [49], and Fuentes et al. [47], in laminar coflow ethylene diffusion flames at normal gravity as well as the present data. The fit is expressed as $(\dot{\omega}_{SF,P} / \dot{\omega}_{SF,P}^0) = (T_{ad} / T_{ad}^0)^m$ with $m=6.17$ and a coefficient of determination, $R^2=0.97$.

cm³/s and 65 L/min, respectively. The air and N₂ flow rates in the oxidizer were adjusted to vary X_{O_2} between 0.209 and 0.152. Fuentes et al. [47] considered both oxygen-enriched and oxygen-reduced laminar coflow ethylene diffusion flames stabilized on a Santoro-type burner with an inner diameter for the central fuel tube of 10.0 mm. The fuel and oxygen flow rates were kept constant at 2.345 cm³/s and 20 L/min, respectively, whereas the nitrogen flow rate was decreased from 97.6 to 37.1 L/min to vary X_{O_2} from 0.17 to 0.35. In their experiments, the SVF was measured by using a laser extinction technique at a wavelength of 670 nm.

SF and SO processes occur mainly in different regions in axisymmetric laminar diffusions flames [48] (see also the discussion in Section 5.4). As a consequence, the peak of SF rate can be estimated from the conservation of soot mass fraction in a Lagrangian sense in the SF region, $\frac{D(\rho_s f_s)}{Dt} \propto \frac{\rho_s f_{s,max}}{\tau_{SF}} = \dot{\omega}_{SF,P}$. For the laminar coflow diffusion flames at normal gravity, τ_{SF} is estimated as $\sqrt{2L_S/a}$, where L_S represents the distance above the burner at which the SVF peaks and $a = g(\frac{T_{ad}}{T_{\infty}} - 1)$ is a buoyant acceleration [50]. As discussed in Section 2, τ_{SF} is computed as L_S/u_{ox} for the present micro-gravity flames (see Table 1). Figure 3 shows that all the experimental data collapse on a single curve when $\dot{\omega}_{SF,P} / \dot{\omega}_{SF,P}^0$ is plotted as a function of T_{ad} / T_{ad}^0 . This correlation includes both the data obtained with constant and varying S , supporting that the adiabatic flame temperature is the predominant parameter affecting the SF process when varying X_{O_2} . These results show also that the exponent m in Eq. (6) can be taken equal to 6.17.

5. Results and discussions

5.1. Computational details

Simulations are performed in an overall computational domain of 4 cm (r) $\times$ 6 cm (z). The computational domain is divided into

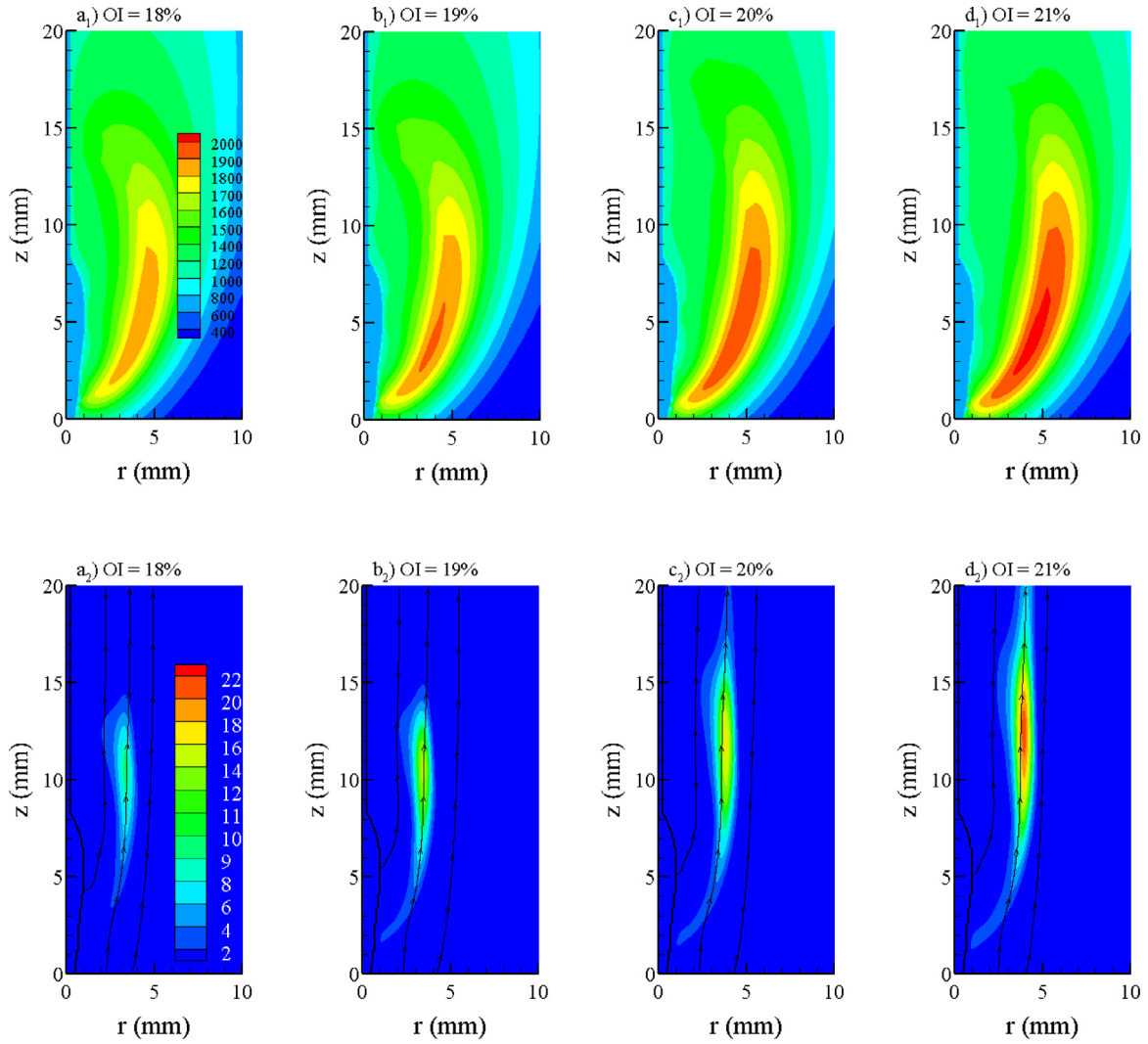


Fig. 4. Computed temperature (index 1, in K) and SVF (index 2, in ppm) fields for a) $X_{O_2} = 18\%$, b) $X_{O_2} = 20\%$, c) $X_{O_2} = 20\%$ and d) $X_{O_2} = 21\%$. Streamlines are plotted with the SVF.

167 (r) $\times$ 378 (z) cells using a non-uniform grid. The finest resolution ($76\mu m \times 76\mu m$) is located in a region covering the molten droplet and the sooting region.

5.2. Flame structure

Figure 4 shows the computed temperature and SVF fields for the four flames. As observed experimentally (see Table 1 and Fig. 2), the predicted flame size and the soot production increase with X_{O_2} . As expected, the temperature peak, that is located at vicinity of the flame leading edge, is also enhanced while increasing X_{O_2} . For a given X_{O_2} , the temperature decreases progressively as the distance from the temperature peak is further increased. This cooling process is related, on the one hand, to the radiative losses and, on the other hand, downstream the molten ball, to the heat sink produced by the bare NiCr.

Figure 4 shows clearly that the cooling in the sooting region is enhanced with X_{O_2} , leading to lower and lower temperature at the flame tip. For $X_{O_2} = 0.20$ and 0.21 , the SO process at the flame tip is not strong enough to fully oxidize soot that, in accordance with the experimental observations for these oxidizer compositions (see Fig. 2), are partially released into the atmosphere. Streamlines are also plotted along with the fields SVF in the bottom diagrams of Fig. 4. It is interesting to note that the model predicts that, what-

ever X_{O_2} , the maximum pathline of SVF coincides approximately with a streamline from $z \approx 5-6$ mm.

5.3. Flame geometry

The objective of this section is to understand the processes that govern the flame geometry and, in turn, the residence times. The analysis assumes complete combustion that is not fully satisfied for the smoking flames ($X_{O_2} \geq 0.20$). However, the amount of smoke released for the present smoking flames does not exceed 5% of the fuel mass flow rate and, therefore, the smoke point transition is expected to have no perceptible influence on the overall trend.

Following the analysis provided by Bhattacharjee et al. for slabs [51] and extended by Konno et al. [52] for cylindrical samples, the luminous flame radius, r_f , is estimated from the oxygen stoichiometric requirement to consume completely the fuel released by pyrolysis:

$$\dot{m}_P \propto \rho_g u_{\alpha} \pi r_f^2 Y_{O_2, \infty} \quad (7)$$

where s is the stoichiometric oxygen to fuel mass ratio. This expression suggests that r_f varies linearly with $\alpha = \sqrt{\dot{m}_P / \rho_g u_{\alpha} Y_{O_2, \infty}}$. In the present spreading configurations, the evolution of α while increasing X_{O_2} is governed by two competitive mechanisms: an increase in $\dot{m}_P$ (see Table 1), that tends to enhance α , and an in-

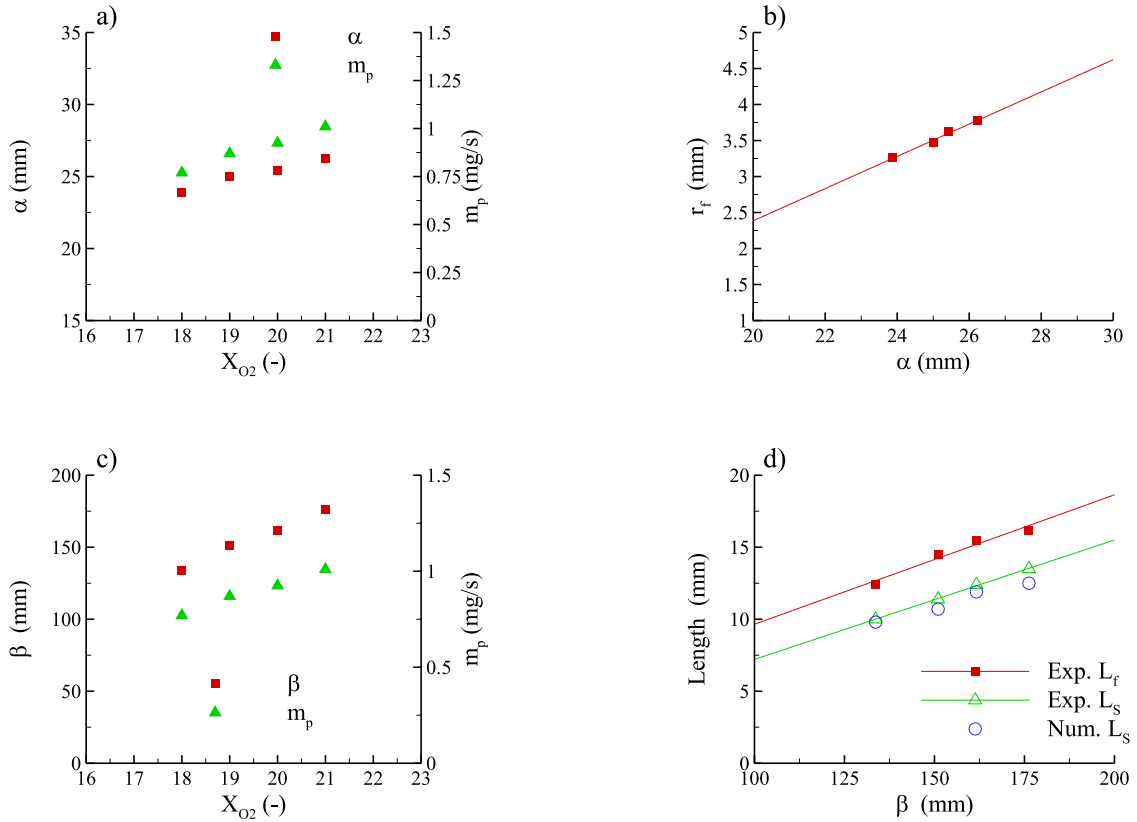


Fig. 5. a) Evolution of α as a function X_{O_2} , b) evolution of experimental r_f as a function of α , c) evolution of β as a function X_{O_2} , and d) evolution of L_f and L_s as a function β .

crease in $Y_{O_{2,\infty}}$, that tends to reduce it. Figure 5a shows that the first mechanism prevails with α increasing with X_{O_2} . Figure 5b confirms the linear dependency of r_f with α .

The analysis of Refs. [51,52] estimates the luminous flame length, L_f , by assuming that the rate of oxygen diffusion towards the fuel equals the stoichiometric amount required for the combustion of the pyrolyzed fuel. In the case of cylindrical samples, this leads to [52]:

$$L_f \propto \beta = \frac{\dot{m}_p \ln(r_f/R_{wire})}{\rho_g \alpha_g Y_{O_{2,\infty}}} \quad (8)$$

where R_{wire} is the radius of the cylindrical wire, i.e. 0.55 mm. Similarly to α , the evolution of β as X_{O_2} is enhanced results from a competition between an enhancement in $\dot{m}_p$, that tends to increase β , and an increase in $Y_{O_{2,\infty}}$, that tends to reduce it. As expected, Figs. 5c and d show that the first mechanism prevails and L_f evolves linearly with β , respectively. Figure 5d shows also that both measured and computed L_s increase linearly with β and the slopes of the linear fits are very similar to that of L_f . These results suggest that the luminous flame length can be alternatively considered as characteristic length scale for the SF process in the present flames. In addition, the computed values of L_s agree well with the experimental ones.

5.4. Soot formation and oxidation processes

Figure 6a shows the SVF along the maximum SVF pathline. It increases in the SF region, reaches a peak, and then decreases owing to the SO process. The numerical model captures well the evolution of SVF in both SF and SO region. As discussed previously and in accordance with experimental observations, the model predicts non-smoking flames for $X_{O_2} = 0.18$ and 0.19 and smoking flames

for $X_{O_2} = 0.20$ and 0.21. In the SF region, the rates of increase in SVF with z as well as the magnitude of the SVF peaks are well reproduced. In the SO region, the rates of decrease in SVF with z are also well reproduced although the model does not predict the abrupt freezing of the SO process as observed experimentally for the smoking flames and, in turn, underestimates the corresponding amount of soot released into the atmosphere. The present oxidation model based on Eqs. (1) and (2) does not consider aging effects that reduce the soot surface reactivity near the flame tip, which may explain these underestimations [53].

Improvements to account for particle aging were reported in the literature in the case of PAH-based soot production models [53,54]. As discussed in Section 4, such advanced soot models are challenging to be applied to solid fuels and are beyond the scope of the present study. The present numerical results suggest that the model captures the main trends observed experimentally and model predictions can be used to complete the present experimental data.

Let us start to discuss the SF region. Figure 6a shows that the peak location of SVF occurs further downstream as X_{O_2} is enhanced, which results in longer residence time to form soot and partially explains the increase in SVF with X_{O_2} . The evolution of $\dot{\omega}_{SF}$ shows that an enhancement in the SF rate also contributes to this increase (see Fig. 6b). Figure 6a shows that the computed temperature in the SF region follows the same trends as the adiabatic flame temperature, increasing with X_{O_2} . This reinforces the assumption made in Section 4.2.2 that thermal effects control the effects of X_{O_2} on the SF rate and can be related empirically to the adiabatic flame temperature.

Model predictions show that the contribution of OH dominates the SO process along the maximum SVF pathline. Figure 6b shows that $\dot{\omega}_{OH}$ exhibits a non-monotonic behavior with X_{O_2} , increasing

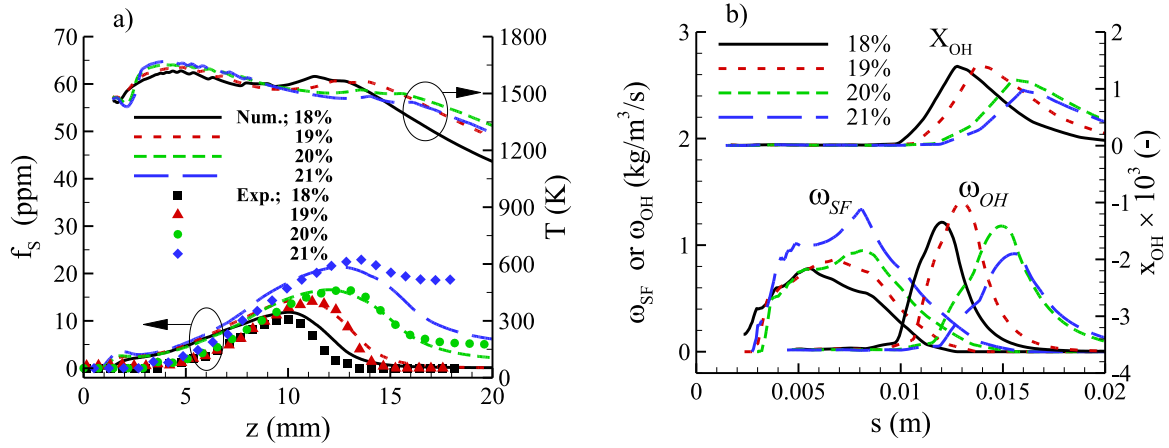


Fig. 6. Evolution along the maximum SVF pathline of a) the experimental and predicted SVF, and the predicted temperature, b) predicted ω_{SF} and ω_{OH} , and OH mole fraction. Experimental data are represented by symbols. Numerical results are represented by lines.

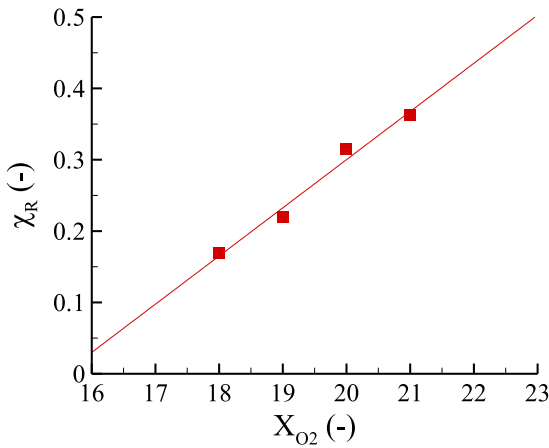


Fig. 7. Computed radiant fraction as a function of X_{O_2} .

first as X_{O_2} is enhanced from 0.18 to 0.19 and, then, decreasing as X_{O_2} is further increased. The previous observations can be then directly related to the effects of increasing X_{O_2} on temperature and OH mole fraction. These effects result from the competition between two mechanisms. On the one hand, oxygen-enhanced oxidizers tend to increase the flame temperature and to promote the formation of OH. On the other hand, Fig. 7 shows that the radiant fraction increases almost linearly with X_{O_2} from about 0.169 for $X_{O_2} = 0.18$ to 0.362 for $X_{O_2} = 0.21$. This increase in radiative losses tends to reduce the temperature in the SO region and to weaken the formation of OH. A detailed examination of Fig. 6 shows that the first mechanism prevails as X_{O_2} is enhanced from 0.18 to 0.19 whereas the second dominates for $X_{O_2} = 0.20$ and 0.21. In particular, it can be observed in Fig. 6 that, in the SO region, the temperature and OH mole fraction become significantly lower and lower as X_{O_2} is increased beyond 0.19.

The experimental SVF in Fig. 6a shows that, for $X_{O_2} = 0.20$ and 0.21, the soot oxidation process freezes at $z = 1.71$ cm and 1.60 cm. At these locations, the corresponding predicted temperatures are of 1451 and 1453 K, respectively. These values stand within the range of 1350–1450 K that corresponds to soot burnout temperature reported in the literature for axisymmetric laminar diffusion flames under different conditions at both normal [34–37] and micro-gravity [24,26]. The smoke point transition occurs for X_{O_2} between 19% and 20%. Figure 7 shows that the corresponding radiant fractions are of 0.23 and 0.30, respectively. These values are

in line with recent measurements at normal gravity, which shows smoke point occurs for radiant fractions of about 0.25 for $X_{O_2} = 19\%$ and 0.28 for $X_{O_2} = 20\%$ [49].

Figure 8 compares measured and predicted radial profiles of SVF at different z . It should be pointed out that a given location does not necessarily correspond to the same stage in the soot production process for the different flames owing to the different residence times. This figure confirms that model predictions reproduce reasonably the experimental trends in each region of the soot production process. Nevertheless, it can be observed that, for a given z , the computed maximum occurs at higher r and the soot region is shifted downstream as compared to the experiments. These discrepancies become more and more pronounced as the distance along the wire axis increases and can be attributed to the fact that soot evolves parallel to the wire in the simulation (see Fig. 4b) whereas it seems to fall down toward the wire at the stage of the flame trailing edge in the experiments (see Fig. 2). It was observed during the experiments that the flame trailing edge is sensitive to small changes in gravity, which may explain at least partially these discrepancies.

5.5. Effects of X_{O_2} on the radiative feedback to the solid surface

Figure 9 shows that, as expected, the radiative heat flux, $\dot{q}''_{R,inc}$, increases substantially with X_{O_2} . Whatever X_{O_2} , $\dot{q}''_{R,inc}$ increases sharply with z along the molten ball, from values around 1–2 kW/m² at the leading edge to reach a peak downstream the ball top. The peak value is strongly enhanced as X_{O_2} increases, from about 9.42 kW/m² for $X_{O_2} = 0.18$ to about 15.39 kW/m² for $X_{O_2} = 0.21$. $\dot{q}''_{R,inc}$ reaches another maximum value along the bare wire in the region where the flame falls back to the wire. This maximum also increases substantially with X_{O_2} . Figure 9 shows also the radiative flux emitted by the wire surface, $\dot{q}''_{R,w}$. It appears clearly that surface radiative losses dominate the flame radiative feedback in the pyrolysis region and along the bare wire. In the pyrolysis region, $\dot{q}''_{R,w}$ is predicted to be constant due to the assumption of constant pyrolysis temperature. Along the bare wire, it increases to reach a maximum in the region where the flame falls back to the wire. This maximum follows a non-monotonic behavior with X_{O_2} , increasing first as X_{O_2} increases from 18% to 20% and then decreasing slightly as X_{O_2} is increased further to 21%. Finally a careful examination of $\dot{q}''_{R,inc}$ and $\dot{q}''_{R,w}$ shows that the contribution of thermal radiation to the heating process of the solid ahead of the pyrolysis front is negligible whatever X_{O_2} .

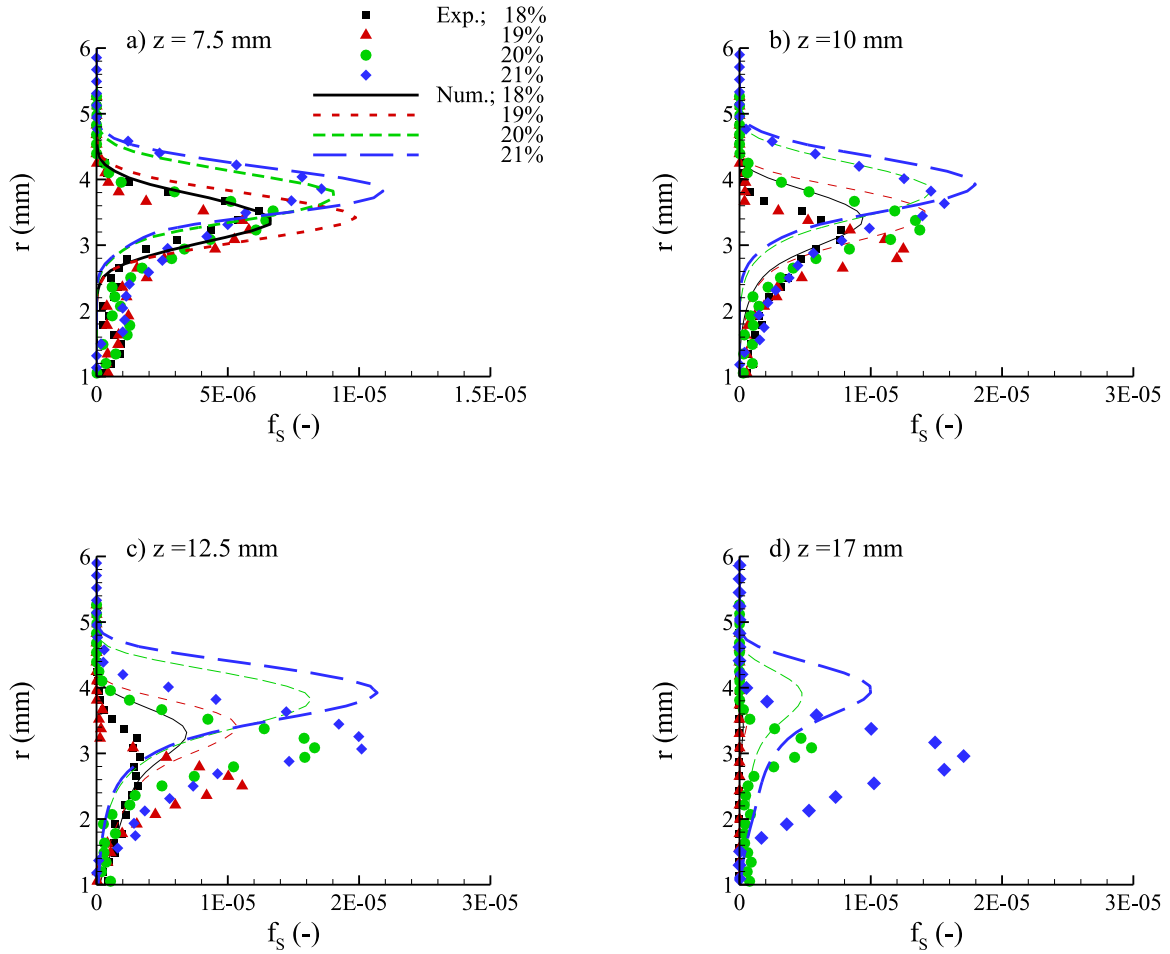


Fig. 8. Radial profiles of SVF at different heights. Experimental data are represented by symbols. Numerical results are represented by lines.

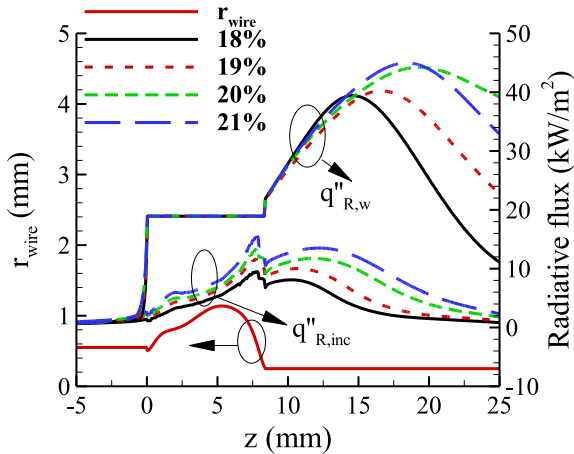


Fig. 9. Incident radiative flux, $\dot{q}''_{R,inc}$, and surface emission flux, $\dot{q}''_{R,w}$, along the wire axis (right y-axis). The axial evolution of the wire radius, r_{wire} , is also plotted (left y-axis).

6. Conclusions

Soot production, smoke emission and radiative heat transfer in oxygen-reduced non-buoyant laminar diffusion flames spreading in opposed flow along NiCr wires coated with LDPE are investigated experimentally and numerically. The numerical model considers a

detailed chemistry, a state-of-the-art radiation model, and a two-equation soot production model involving a smoke-point soot formation rate and oxidation by OH and O₂.

The following conclusions can be drawn:

- 1) Experimental data show that spread rate, fuel mass flow rate, flame size, residence time and peak of soot volume fraction increase with X_{O_2} . The flame length and radius and, in turn, the residence time increase with X_{O_2} owing to the increase in the fuel mass flow rates.
- 2) Analysis of experimental data shows that increasing X_{O_2} enhances the soot formation rate through thermal effects that can be related to the adiabatic flame temperature. Based on these experimental observations, the soot formation rate was scaled by the adiabatic flame temperature to account for X_{O_2} effects.
- 3) Model predictions are in quantitative agreement with the available experimental data, capturing the effects of X_{O_2} on flame structure, soot formation and oxidation processes. In addition, the model reproduces well the transition from a non-smoking to a smoking flame that occurs for X_{O_2} between 19% and 20%. These results support that the proposed surrogate fuel modeling strategy and the extended soot production model are promising to be applied to other configurations involving solid fuels at both normal and micro-gravity.
- 4) Increasing X_{O_2} affects the soot oxidation process through two competitive mechanisms. On the one hand, oxygen-enhanced conditions promote the formation of soot oxidizing species. On the other hand, this increases the radiative losses which tends

to lower the temperature and weaken the formation of these species in the soot oxidation region. For the present flames, the first mechanism dominates as X_{O_2} is increased from 18% to 19% whereas the second prevails as X_{O_2} is further increased beyond 19%.

- 5) The radiant fraction increases with X_{O_2} to reach about 25%–30% at the SP transition. In addition, for the smoking flames, the soot oxidation freezing temperature is found to be about 1450 K. These SP characteristics in terms of radiant fraction and soot oxidation freezing temperature are consistent with those previously reported at normal gravity.
- 6) The net radiative flux ahead of the pyrolysis front is close to zero whatever X_{O_2} , showing that radiation does not participate to the heating process responsible for flame spread. On the other hand, surface radiative losses dominate the radiative feedback from the flame along the pyrolysis region and the bare wire.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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